

DETERMINANT

Determinant:

- A Determinant is a number associated with a matrix. Only SQUARE matrices have a determinant.
- The symbol for a determinant can be the phrase “det” in front of a matrix variable, $\det(A)$; or vertical bars around a matrix, $|A|$

It can be expressed as,

$$C = [4]_{1 \times 1} \quad |C| = 4$$

$$C = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}_{2 \times 2} \quad |C| = (a_1 * b_2) - (b_1 * a_2)$$

$$C = \begin{bmatrix} 3 & 2 \\ 5 & 1 \end{bmatrix}_{2 \times 2} \quad |C| = (3 * 1) - (2 * 5) = 3 - 10 = -7$$

Minors and Cofactors:

- If A is a square matrix, then the **minor of entry a_{ij}** is denoted by M_{ij} and is defined to be the determinant of the sub-matrix that remains after the i th row and j th column are deleted from A . The number $(-1)^{i+j}M_{ij}$ is denoted by C_{ij} and is called the **cofactor of entry a_{ij}** .

Example: A matrix B is given

$$B = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

The minor of the entry a_{11} is

$$M_{11} = \begin{vmatrix} \cancel{a_{11}} & \cancel{a_{12}} & \cancel{a_{13}} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$$

The cofactor of the entry a_{11} is

$$C_{11} = (-1)^{1+1} M_{11} = (-1)^{1+1} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$$

Problem 01: Finding Minors and Cofactors of the elements a_{11} and a_{32} of the following matrix.

$$B = \begin{bmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{bmatrix}$$

Solution: The minor of the entry a_{11} is

$$M_{11} = \begin{vmatrix} \cancel{3} & \cancel{1} & \cancel{-4} \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{vmatrix} = \begin{vmatrix} 5 & 6 \\ 4 & 8 \end{vmatrix} = 5 \times 8 - 4 \times 6 = 40 - 24 = 16$$

The cofactor of the entry a_{11} is

$$C_{11} = (-1)^{1+1} M_{11} = (-1)^{1+1} \begin{vmatrix} 5 & 6 \\ 4 & 8 \end{vmatrix} = (-1)^2 (5 \times 8 - 4 \times 6) = 40 - 24 = 16$$

The minor of the entry a_{32} is

$$M_{32} = \begin{vmatrix} 3 & \cancel{1} & -4 \\ 2 & 5 & 6 \\ \cancel{1} & \cancel{4} & \cancel{8} \end{vmatrix} = \begin{vmatrix} 3 & -4 \\ 2 & 6 \end{vmatrix} = 3 \times 6 - (-4) \times 2 = 18 + 8 = 26$$

The cofactor of the entry a_{32} is

$$C_{32} = (-1)^{3+2} M_{32} = (-1)^{3+2} \begin{vmatrix} 3 & -4 \\ 2 & 6 \end{vmatrix} = (-1)^5 \{3 \times 6 - (-4) \times 2\} = -(18 + 8) = -26.$$

Determinant (Using cofactor expansion):

If A is an $n \times n$ matrix, then the number obtained by multiplying the entries in any row or column of A by the corresponding cofactors and adding the resulting products is called the **determinant of A** , and the sums themselves are called **cofactor expansions of A** . That is,

$$\det(A) = a_{1j}C_{1j} + a_{2j}C_{2j} + \dots + a_{nj}C_{nj}$$

[Cofactor expansion along the j th column]

$$\det(A) = a_{i1}C_{i1} + a_{i2}C_{i2} + \dots + a_{in}C_{in}$$

[Cofactor expansion along the i th row]

Problem 02: Find the determinant of the matrix

$$B = \begin{bmatrix} 3 & 1 & 0 \\ -2 & -4 & 3 \\ 5 & 4 & -2 \end{bmatrix}$$

by cofactor expansion along the first row.

Solution:

$$\det(B) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$$

$$\Rightarrow \det(B) = 3 \cdot (-1)^{1+1} \cdot \begin{vmatrix} -4 & 3 \\ 4 & -2 \end{vmatrix} + 1 \cdot (-1)^{1+2} \cdot \begin{vmatrix} -2 & 3 \\ 5 & -2 \end{vmatrix} + 0 \cdot (-1)^{1+3} \cdot \begin{vmatrix} -2 & -4 \\ 5 & 4 \end{vmatrix}$$

$$\Rightarrow \det(B) = 3(8-12) - 1(4-15) + 0 = 3(-4) - (-11) = -12 + 11 = -1$$

Determinant of a triangular (upper or lower) matrix or diagonal matrix:

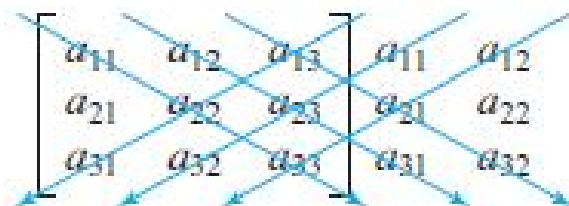
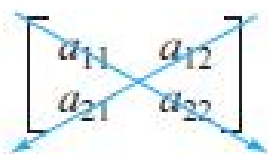
If A is an $n \times n$ triangular (upper triangular, lower triangular) or diagonal matrix, then $\det(A)$ is the product of the entries on the main diagonal of the matrix; that is, $\det(A) = a_{11}a_{22}a_{33}\dots a_{nn}$.

Example: Let

$$B = \begin{bmatrix} 3 & 0 & 0 \\ -2 & -4 & 0 \\ 5 & 4 & -2 \end{bmatrix}$$

be a matrix, then the determinant is $\det(A) = 3 \times (-4) \times (-2) = 24$.

Arrow technique to evaluate the determinant:



That's mean, $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \Rightarrow (a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}) - (a_{13}a_{22}a_{31} + a_{11}a_{23}a_{32} + a_{12}a_{21}a_{33})$

Example: Let a matrix defined by

$$B = \begin{bmatrix} 3 & 1 & 0 \\ -2 & -4 & 3 \\ 5 & 4 & -2 \end{bmatrix}$$

then the determinant by using arrow technique is

$$\begin{aligned} \det(B) &= (a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}) - (a_{13}a_{22}a_{31} + a_{11}a_{23}a_{32} + a_{12}a_{21}a_{33}) \\ \Rightarrow \det(B) &= \{3 \cdot (-4) \cdot (-2) + 1 \cdot 3 \cdot 5 + 0 \cdot (-2) \cdot 4\} - \{0 \cdot (-4) \cdot 5 + 3 \cdot 3 \cdot 4 + 1 \cdot (-2) \cdot (-2)\} \\ \Rightarrow \det(B) &= (24 + 15 + 0) - (0 + 36 + 4) = 39 - 40 = -1 \end{aligned}$$

Properties of Determinants of Matrices:

1. Determinant evaluated across any row or column is **same**.
2. If all the elements of a row (or column) are zeros, then the value of the determinant is **zero**.
3. Determinant of an Identity matrix (I_n) is **1**.
4. If rows and columns are interchanged then value of determinant remains same (value does not change). Therefore, **$\det(A) = \det(A^T)$** , here A^T is transpose of matrix A.
5. If any two row (or two column) of a determinant are interchanged the value of the determinant is multiplied by **-1**.
6. If all elements of a row (or column) of a determinant are multiplied by some scalar number **k**, the value of the new determinant is **k** times of the given determinant. Therefore, if **A** be an n-rowed square matrix and K be any scalar. Then **$|KA| = K^n|A|$** .
7. If two rows (or columns) of a determinant are **identical (same)** the value of the determinant is **zero**.
8. Let A and B be two matrix, then **$\det(AB) = \det(A) \cdot \det(B)$** .
9. If A be a matrix then, **$|A^n| = |A|^n$** .
10. Determinant of Inverse of matrix can be defined as **$|A^{-1}| = |A|^{-1}$** .
11. Determinant of diagonal matrix, triangular matrix (upper triangular or lower triangular matrix) is product of element of the principal diagonal.
12. In a determinant each element in any row (or column) consists of the sum of two terms, then the determinant can be expressed as sum of two determinants of same order. For example,

$$\begin{vmatrix} a1 + x1 & b1 & c1 \\ a2 + x2 & b2 & c2 \\ a3 + x3 & b3 & c3 \end{vmatrix} = \begin{vmatrix} a1 & b1 & c1 \\ a2 & b2 & c2 \\ a3 & b3 & c3 \end{vmatrix} + \begin{vmatrix} x1 & b1 & c1 \\ x2 & b2 & c2 \\ x3 & b3 & c3 \end{vmatrix}$$

13. If B is obtained by adding c-times a row of A to a different row, the **det(B) = det(A)**.

14. If value of determinant Δ becomes zero by substituting $x = \alpha$, then $x - \alpha$ is a factor of α .

15. Here, c_{ij} denotes the cofactor of elements of a_{ij} in Δ .

$$\Delta = \begin{vmatrix} a11 & a12 & a13 \\ a21 & a22 & a23 \\ a31 & a32 & a33 \end{vmatrix} \quad \Delta_1 = \begin{vmatrix} c11 & c12 & c13 \\ c21 & c22 & c23 \\ c31 & c32 & c33 \end{vmatrix} = \Delta^2$$

16. In a determinant the sum of the product of the elements of any row (or column) with the cofactors of the corresponding elements of any other row (or column) is zero.

For example, $\det(A) = a_{i1}C_{j1} + a_{i2}C_{j2} + a_{i3}C_{j3} + \dots + a_{in}C_{jn}$, here $C_{j1}, C_{j2}, C_{j3} \dots C_{jn}$ are cofactors along elements of j th row.

17. Let $\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_n$ are the Eigenvalues of A (square matrix of order n). Then **det(A)** = $\lambda_1 \lambda_2 \lambda_3 \dots \lambda_n$, product of Eigenvalues.

DIFFERENCE BETWEEN MATRIX AND DETERMINANT:

Matrix	Determinant
A matrix cannot be reduced to a single number.	A determinant can be reduced to a single number
In a matrix, the number of rows may not be equal to the number of columns.	In a determinant, the number of rows must be equal to the number of columns.
An interchange of rows or columns gives a different matrix.	An interchange of rows or columns gives the same determinant with +ve or -ve sign.
Example: $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$	Example: $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$

N.B:

Let M be a matrix & $|M|$ be the corresponding determinant.

Rectangular	↔	$ M $ undefined
Singular	↔	$ M = 0$
Nonsingular	↔	$ M \neq 0$

Exercise

1. Find the following minors and cofactors of the matrix A and B .

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 6 & 7 & -1 \\ -3 & 1 & 4 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 3 & 6 \\ 0 & 1 & 4 \end{bmatrix}$$

Find,

- | | |
|--------------------------|--------------------------|
| a) M_{13} and C_{13} | b) M_{23} and C_{23} |
| c) M_{22} and C_{22} | d) M_{21} and C_{21} |

Also find the determinants.

2. Find the following minors and cofactors of the matrix A and B .

$$A = \begin{bmatrix} 4 & -1 & 1 & 6 \\ 0 & 0 & -3 & 3 \\ 4 & 1 & 0 & 14 \\ 4 & 1 & 3 & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 3 & -1 & 1 \\ -3 & 2 & 0 & 3 \\ 3 & -2 & 1 & 0 \\ 3 & -2 & 1 & 4 \end{bmatrix}$$

Find,

- | | |
|--------------------------|--------------------------|
| a) M_{32} and C_{32} | b) M_{44} and C_{44} |
| c) M_{41} and C_{41} | d) M_{24} and C_{24} |

Also find the determinants.

3. Use the arrow technique to evaluate the determinants of the following matrices.

a. $\begin{bmatrix} a-3 & 5 \\ -3 & a-2 \end{bmatrix}$

b. $\begin{bmatrix} -2 & 7 & 6 \\ 5 & 1 & -2 \\ 3 & 8 & 4 \end{bmatrix}$

c. $\begin{bmatrix} -1 & 1 & 2 \\ 3 & 0 & -5 \\ 1 & 7 & 2 \end{bmatrix}$

d. $\begin{bmatrix} c & -4 & 3 \\ 2 & 1 & c^2 \\ 4 & c-1 & 2 \end{bmatrix}$

4. Use cofactor expansion along a row or column of your choice to evaluate the determinants of the following matrices.

i. $\begin{bmatrix} -2 & 7 & 6 \\ 5 & 1 & -2 \\ 3 & 8 & 4 \end{bmatrix}$

ii. $\begin{bmatrix} 1 & k & k^2 \\ 1 & k & k^2 \\ 1 & k & k^2 \end{bmatrix}$

iii. $\begin{bmatrix} k+1 & k-1 & 7 \\ 2 & k-3 & 4 \\ 5 & k+1 & k \end{bmatrix}$

iv. $\begin{bmatrix} 3 & 3 & 0 & 5 \\ 2 & 2 & 0 & -2 \\ 4 & 1 & -3 & 0 \\ 2 & 10 & 3 & 2 \end{bmatrix}$

v. $\begin{bmatrix} 4 & 0 & 0 & 1 & 0 \\ 3 & 3 & 3 & -1 & 0 \\ 1 & 2 & 4 & 2 & 3 \\ 9 & 4 & 6 & 2 & 3 \\ 2 & 2 & 4 & 2 & 3 \end{bmatrix}$

5. Evaluate the determinant of the following matrix by a cofactor expansion along,

$$\begin{bmatrix} 3 & 0 & 0 \\ 2 & -1 & 5 \\ 1 & 9 & -4 \end{bmatrix}$$

(a) the first row.

(b) the first column.

(c) the second row.

(d) the second column.

(e) the third row.

(f) the third column.